# Section 1
# Neuroanatomy

Sujin Park

COGS 17

10/03/25

UC San Diego

# Sujin Park

Education:
2nd year PhD Student in Cognitive Science
MA Psychology (Cognitive Neuroscience)
BA Political Science and Diplomacy & Psychology

Research:
- Biomarkers of Neurodevelopmental disorders

Contact Info:
- Email: sup031@ucsd.edu (Pls include COGS17 in subject line) or Canvas Inbox
- Discussion Section: Friday 9 & 10am (online)
- Office Hours: Mon 3:30-4:30pm (zoom link on the announcement on canvas)

Feel free to reach out if you have any questions or problems!

# Ground Rules

**Use this section to boost your learning**

- The aim of this section is to review contents covered in class
- Attending section and actively engaging will improve your learning!
- It's okay to be wrong

**Keep discussions on topic**

- Everyone has different opinions about various things
- Let's keep the conversation about class and lecture subjects

Check these out:

- **Section slides are on github (scan QR code)**
- Scribblesheet: 30s recap after the section where you can put your random thoughts and e-meet your classmates

**Let's all get As!**

# Important Reminders

**Homework Problem sets**

- Homework problems are <u>Required</u>, they guide your learning and will inform us of how you are doing for the lectures
- Due every <u>Friday</u> at Midnight (**EXCEPT** Oct 15, Nov 26)
- <u>IMPORTANT: NO LATE HOMEWORKS WILL BE ACCEPTED</u>

**Exams**

- 4 Exams total: Online, Open book, "one shot" for consecutive 80 Minutes
- 3 Midterms are NON-cumulative
- 1 Final is Comprehensive (on the SAME DAY after 3rd Midterm)

**Extra Credit (Max 20 points)**

- SONA
- Mnemonics
- Homeworks

# Anatomy and Function of the Brain

Nervous System

Cells

Vision, Audition, other sensory systems

Movement

Sleep, Sexual Behavior

Learning

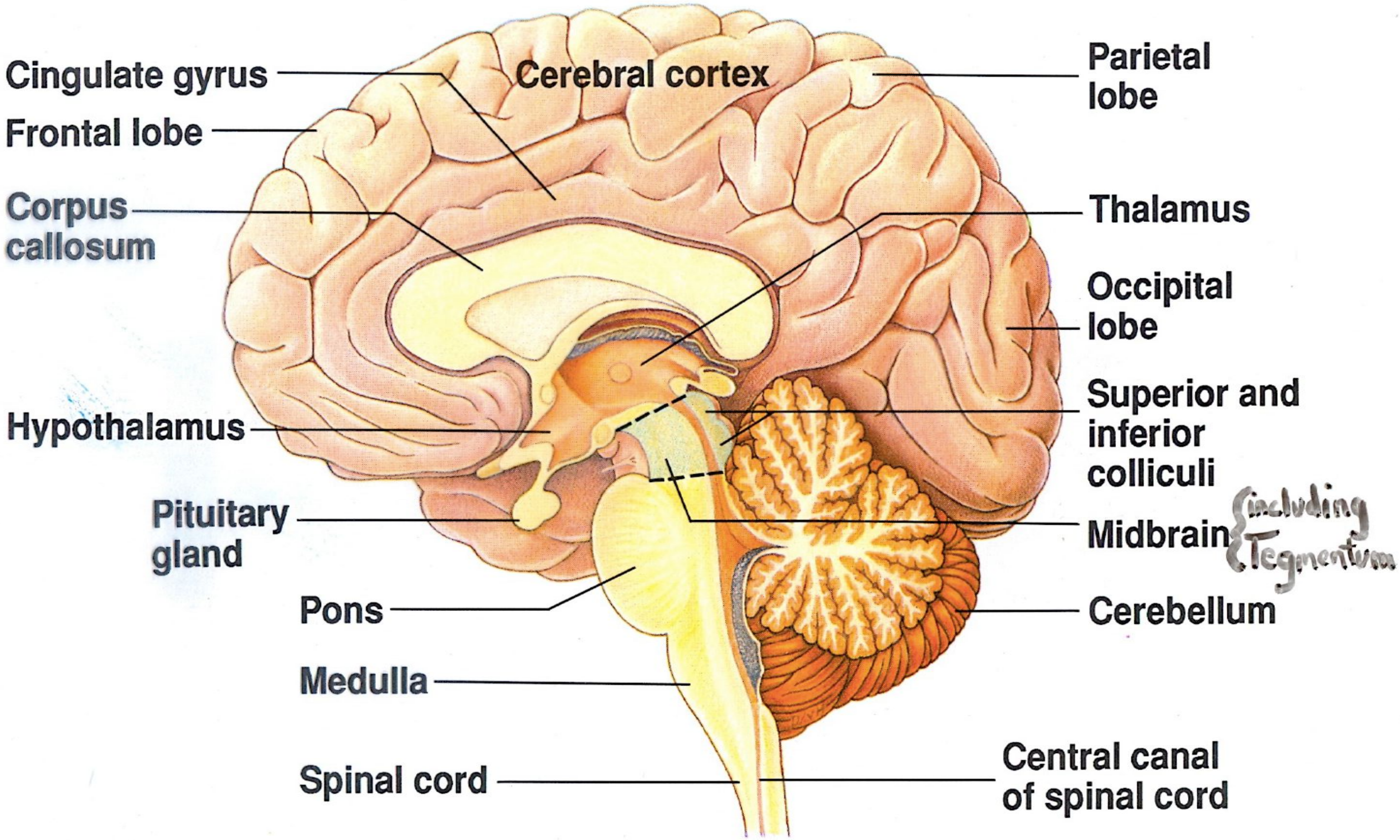
Language

…

# Lecture 1

# Anatomy of the Nervous System

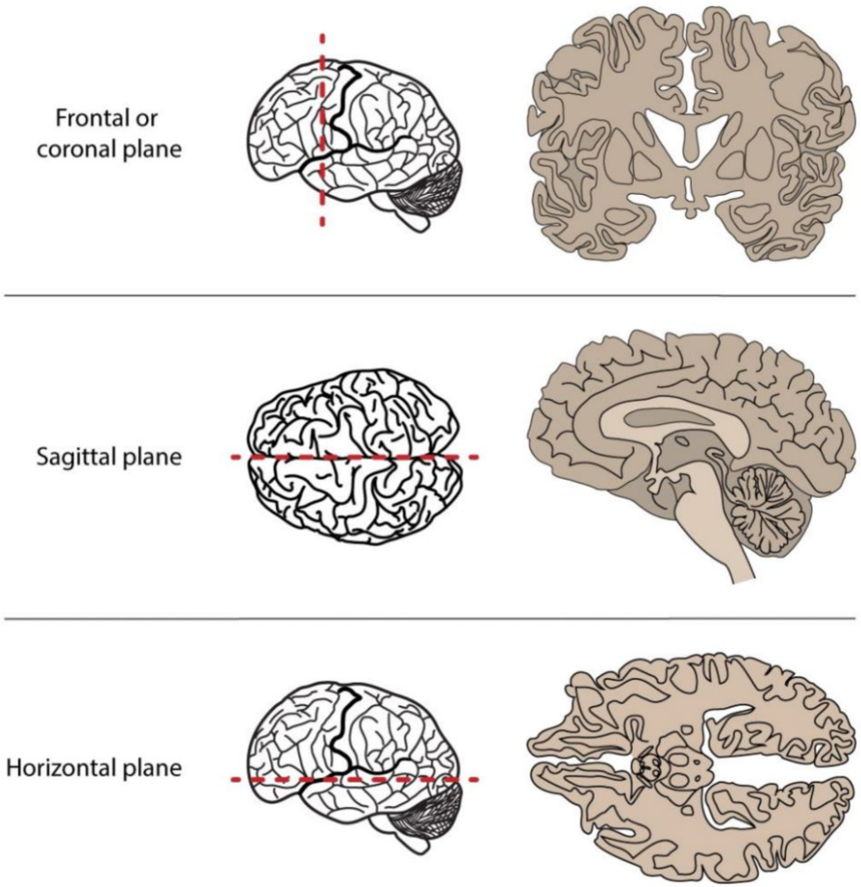
# Mid-Saggital Section




*Sagittal section of human brain*

After Nieuwenhuys et al., 1988

# Planar Views of the Brain

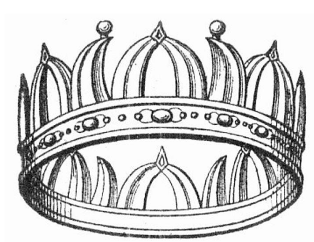



Coronal Plane -- From the **FRONT**

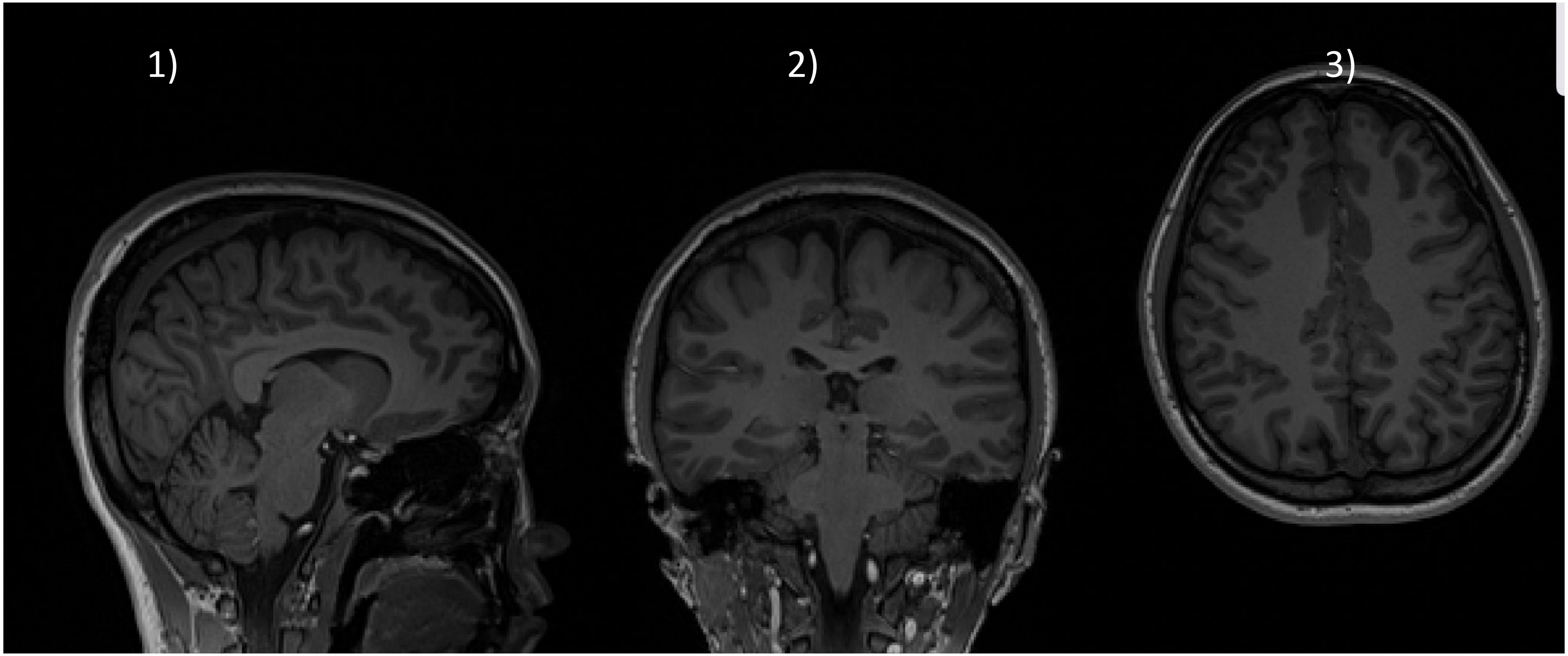
Sagittal Plane -- From the **SIDE**

Horizontal Plane -- From the **ABOVE**

*Corona navalis* (naval crown)

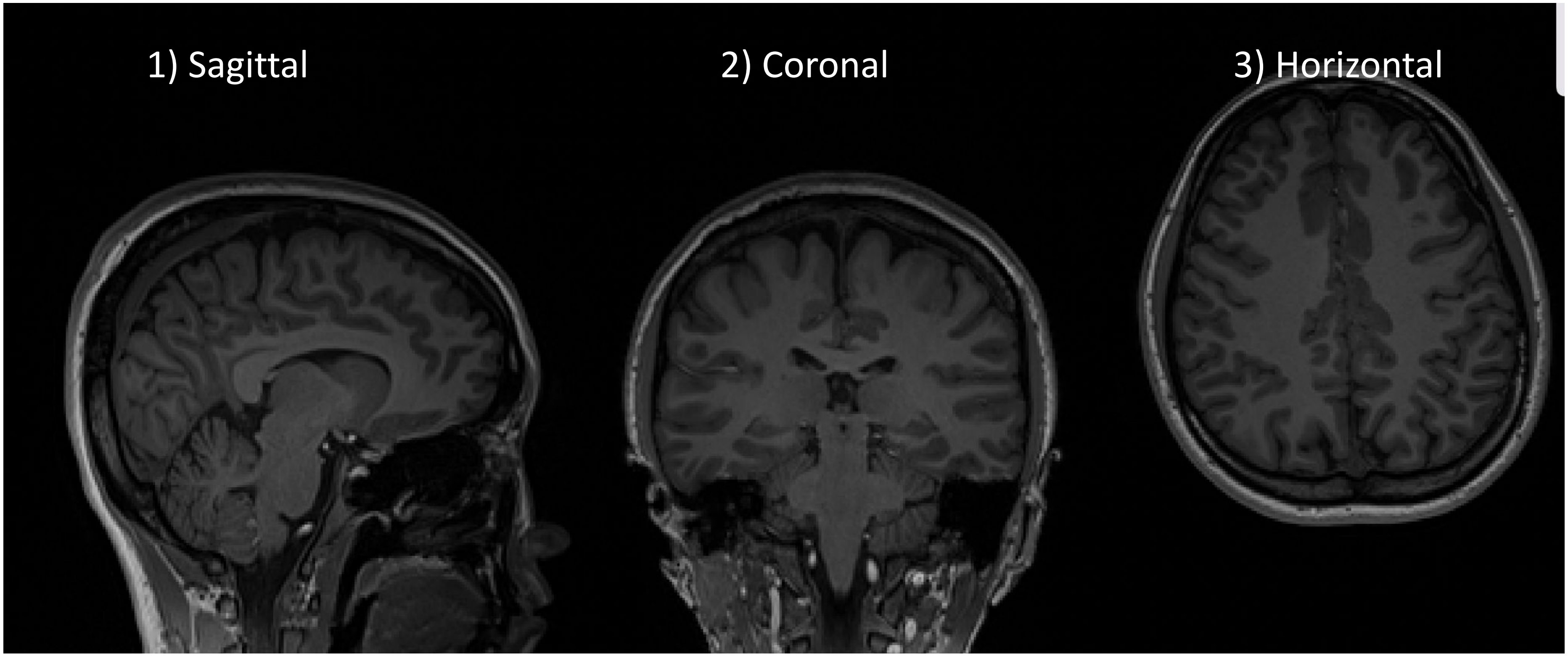


# Planar Views of the Brain




Guess whose brain this is 😏

# Planar Views of the Brain




Guess whose brain this is 😏

# Orientation and Views

**Lateral & Medial**
- Lateral: Towards the sides (Outside)
- Medial: Towards the middle (Center)

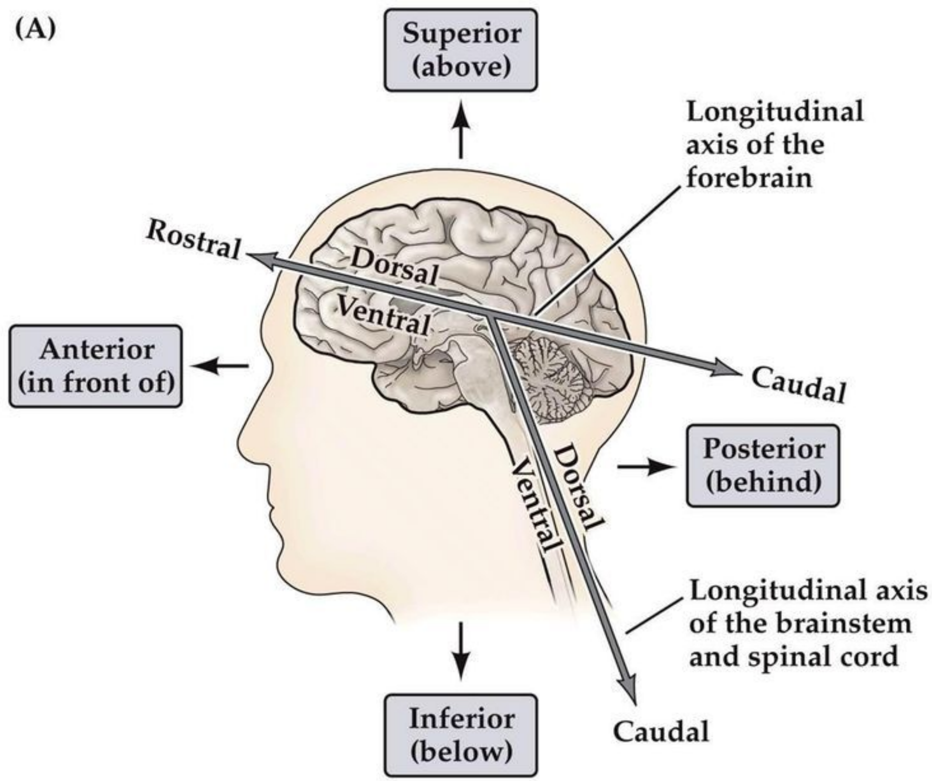
**Dorsal & Ventral**
- Dorsal: The "top" of the brain
- Ventral: The "underside" of the brain

* note dorsal = "rear", ventral = "front" for the body

**Anterior & Posterior**
- Anterior: Front
- Posterior: Back

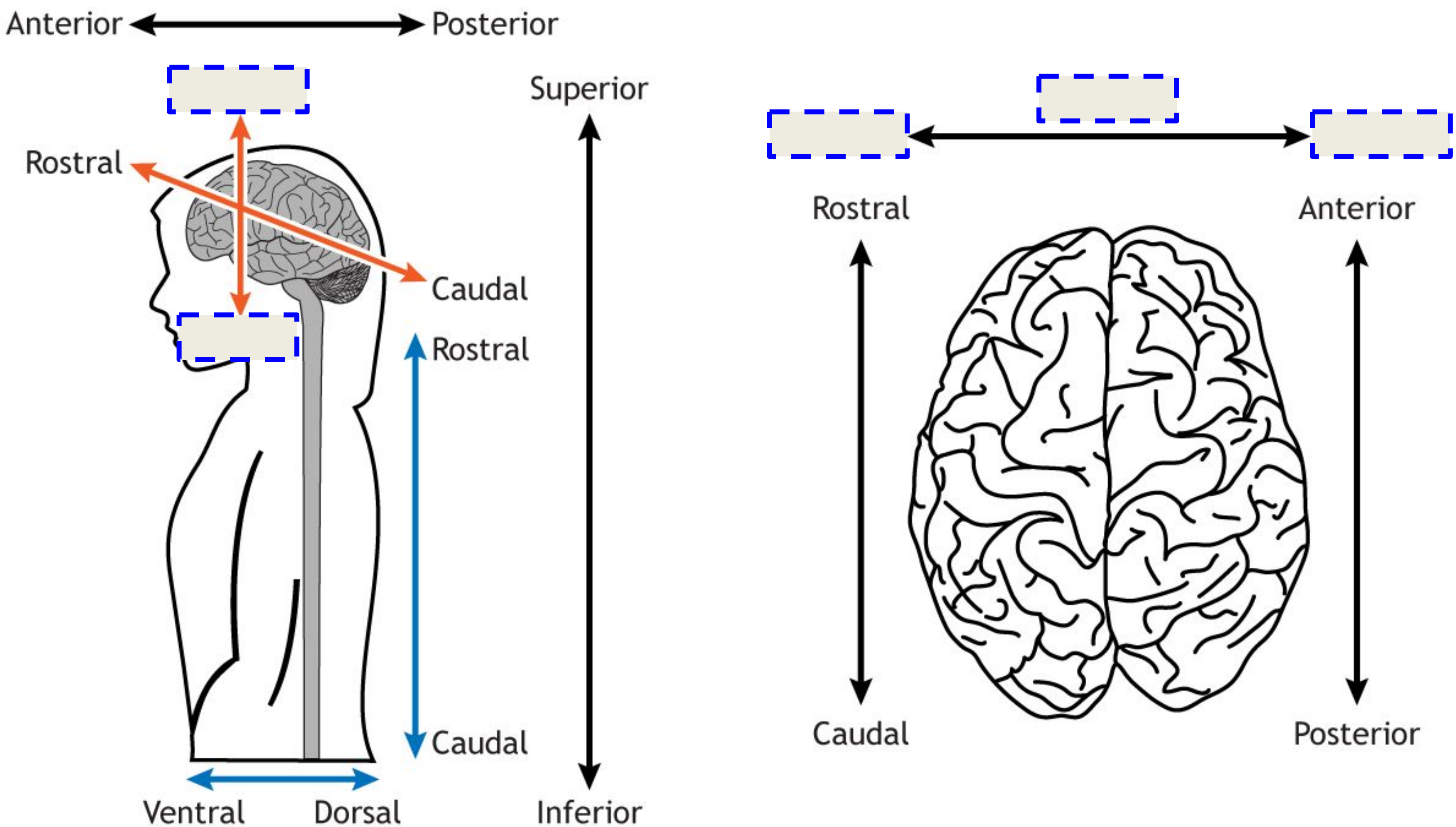
**Bilateral Structure**
- Ipsilateral: Same side
- Contralateral: Opposite side

e.g., Left motor cortex controls right side of the body



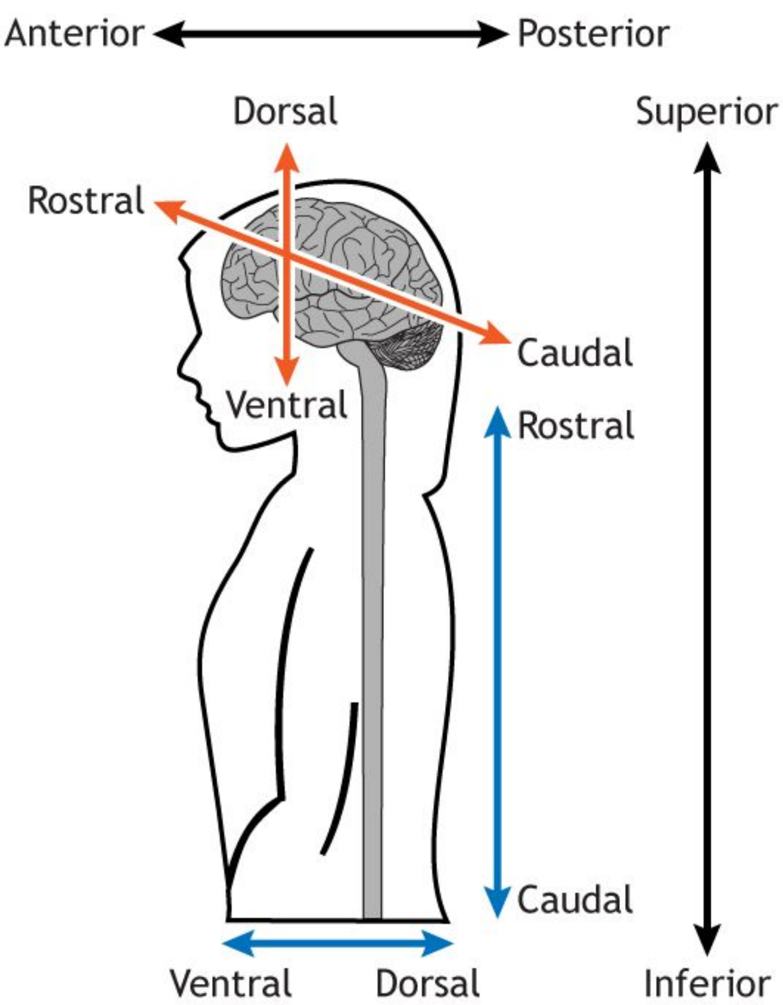

*NEUROSCIENCE 5e*, Figure A1 (Part 1)

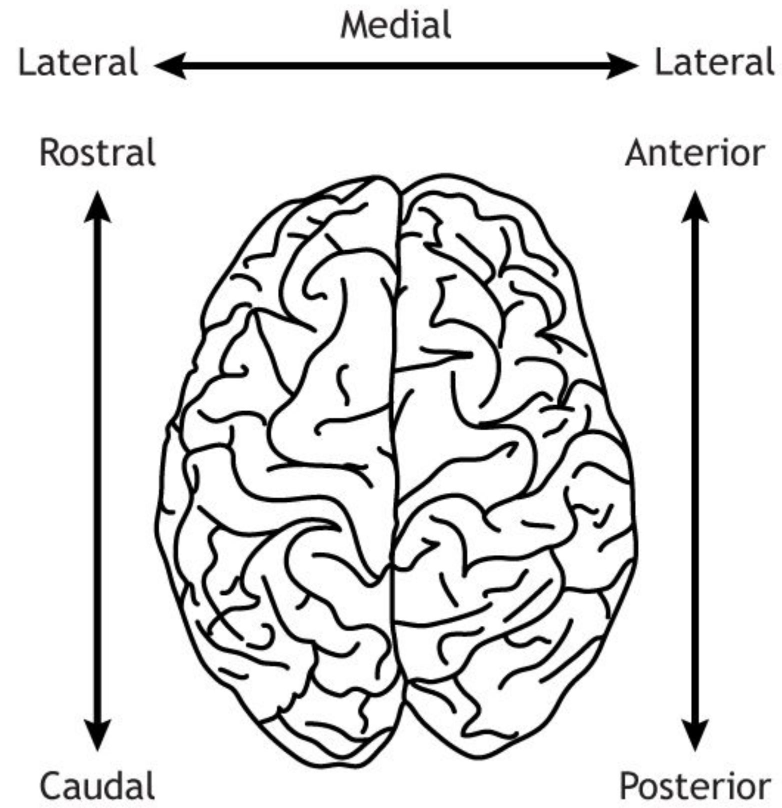
# Orientation and Views




https://openbooks.lib.msu.edu/neuroscience/chapter/anatomical-terminology/

# Orientation and Views

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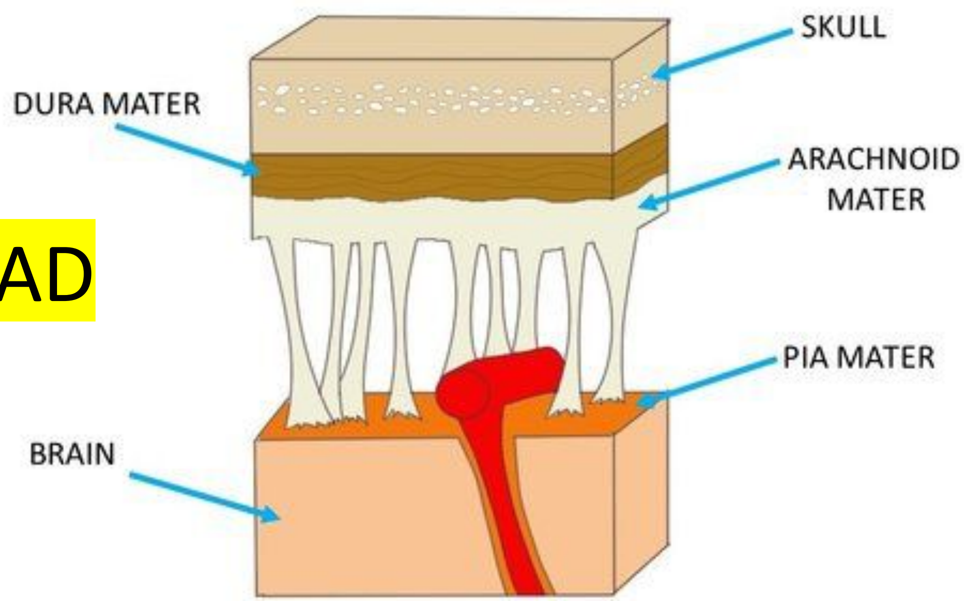
# Support Structure

**Meninges:** surrounds CNS under bone with 3 main layers

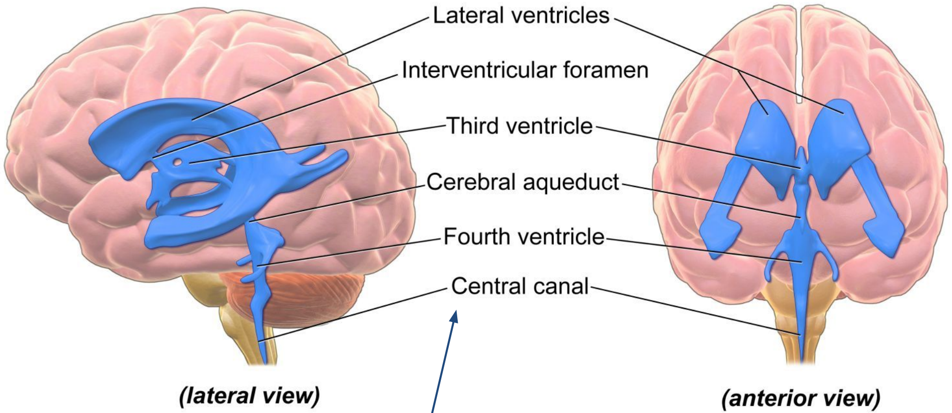
- Dura Mater (=Tough mother): Thick outer layer
- Arachnoid Mater: Spider-web like, spongy layer filled with Cerebrospinal fluid (CSF), shock absorber
- Pia Mater: flexible inner layer that conforms to the brain and spine surfaces, include blood vessels




PAD

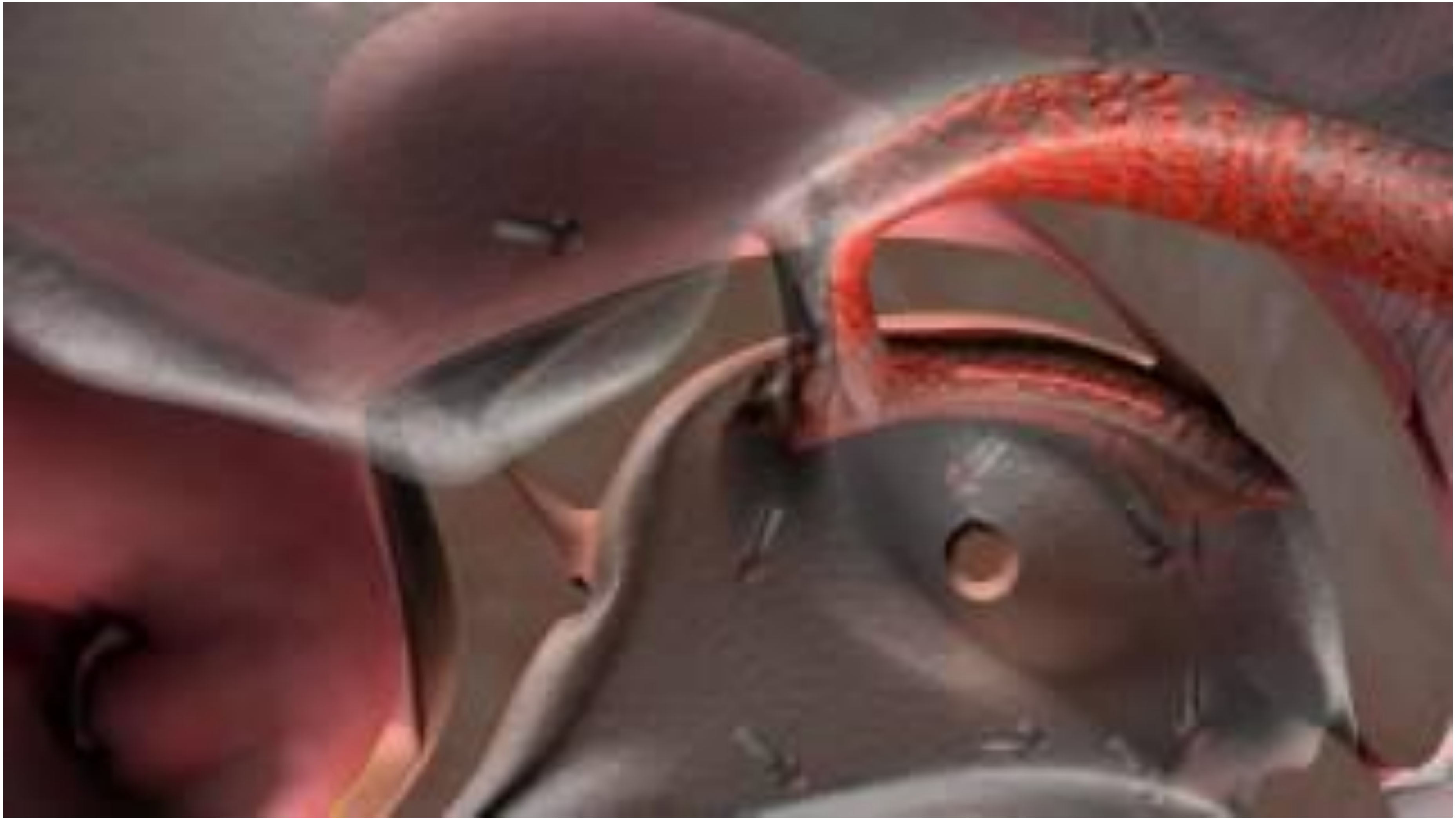
## Ventricles

- Hollow (but not empty!), interconnected cavities
- produce and filled with CSF (also circulate it)
- Structure:
  - 2 Lateral Ventricles
  - Central Third Ventricle
  - Cerebral Aqueduct
  - Central Fourth Ventricle




only structure that is not duplicated on either side of the brain

# Support Structure

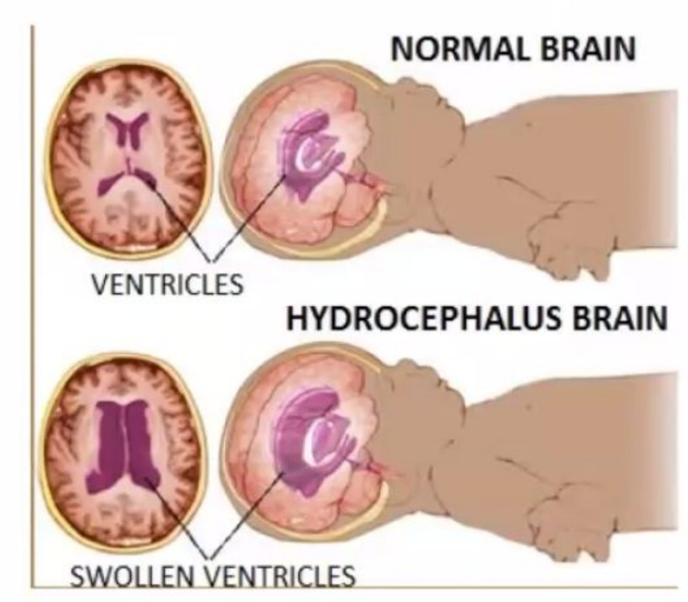


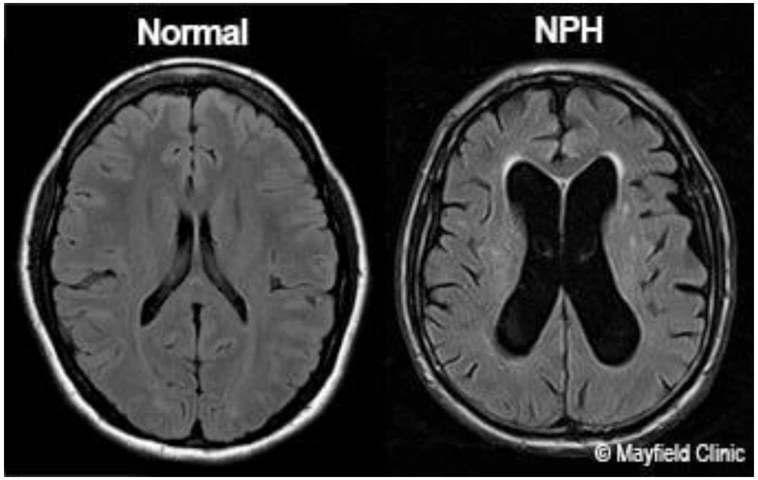
## Swollen ventricles

- When does this happen: CSF is not properly drained out through the cerebral aqueduct, the ventricles tend to swell up

- Then… CSF swelling pushes the brain matter against the PAD, replacing cortical matter with CSF

- Typically Fatal
    - Interventions can redirect excess CSF into the abdominal cavity to reduce the swelling



# Hydrocephalus

# Feeding the brain

## Blood Vessels

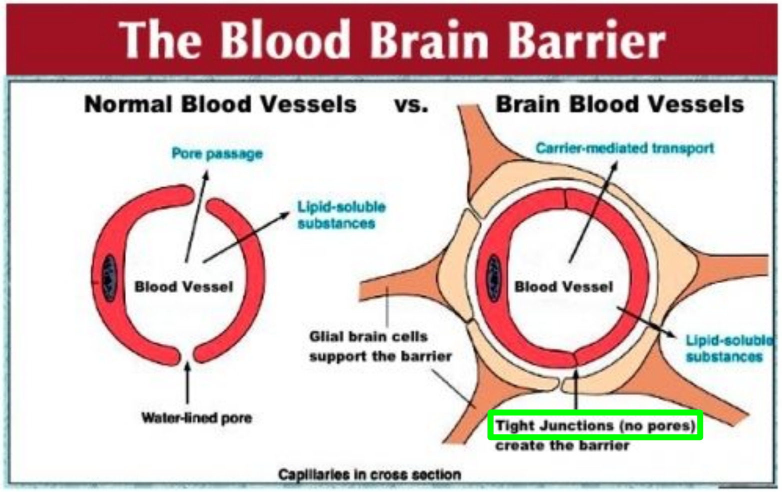
- Web of incoming **arteries** and outgoing **veins** (together, capillaires)
- Helps clear out the brain of waste
- Carries out “used” CSF
- 2% of body weight but uses 20% of blood supply

## Blood-Brain Barrier (BBB)

- Strict control over what enters brain from bloodstream (x free flow!)
- Protects the brain from infections

- Only small uncharged particles (O2, CO2) and some fat-soluble molecules can passively cross BBB

- Astrocytes (Type of Glia cells) works as further barrier and actively fetch whatever needed and put them in the blood vessel (e.g., sugar)



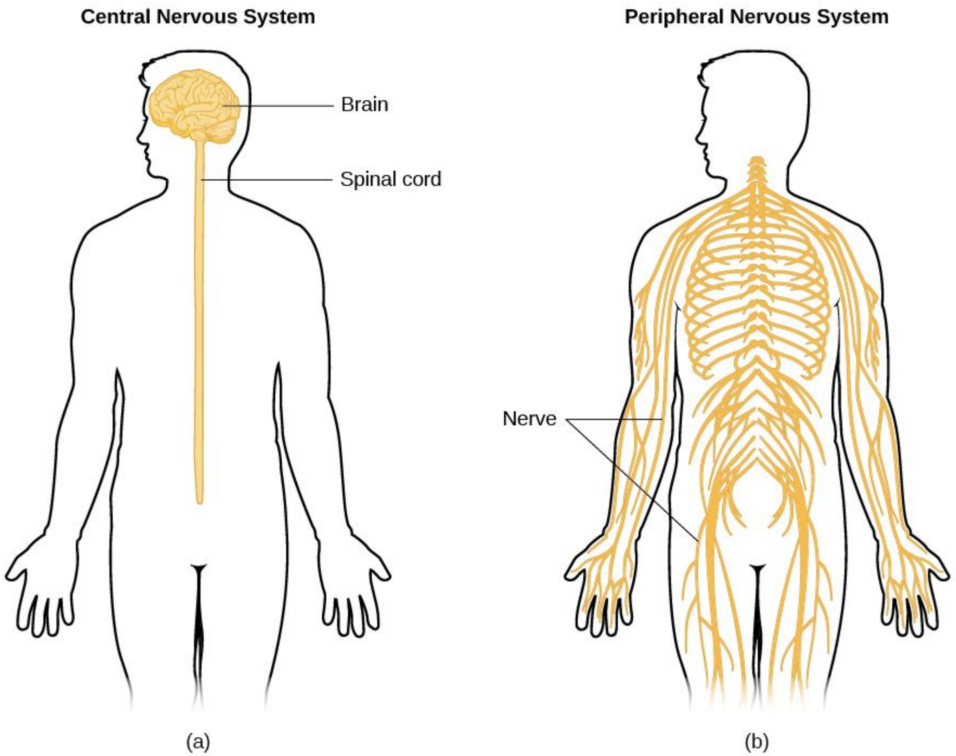
# CNS vs PNS

**Central Nervous System (CNS)**

- Spinal cord and brain
- Encased in bone and meninges

**Peripheral Nervous System (PNS)**

- Nerves outside the CNS
- Somatic Nervous System: interaction with the external environment (Sensory/Motor)
- Autonomic Nervous System: interaction with the internal environment (internal organs)



# Review Questions

*Match definition to concept:*

| | |
|---|---|
| ______ Towards the center of the brain | A. Lateral |
| ______ Bottom of the brain | B. Medial |
| ______ Top of the brain | C. Dorsal |
| ______ On both sides of the brain | D. Ventral |
| ______ Same-side connections | E. Bilateral |
| ______ Towards the outside of the brain | F. Ipsilateral |
| ______ Opposite side connections | G. Contralateral |

*Label as either pertaining to the central nervous system (CNS) or peripheral nervous system (PNS)*:

| | |
|---|---|
| ______ Somatic nervous system | ______ Surrounded by bone and meninges |
| ______ Brain and spinal cord | ______ Autonomic nervous system |

# Review Questions

*Match definition to concept:*

| | | |
|---|---|---|
| B | Towards the center of the brain | A. Lateral |
| D | Bottom of the brain | B. Medial |
| C | Top of the brain | C. Dorsal |
| E | On both sides of the brain | D. Ventral |
| F | Same-side connections | E. Bilateral |
| A | Towards the outside of the brain | F. Ipsilateral |
| G | Opposite side connections | G. Contralateral |

*Label as either pertaining to the central nervous system (CNS) or peripheral nervous system (PNS)*:

PNS Somatic nervous system

CNS Brain and spinal cord

CNS Surrounded by bone and meninges

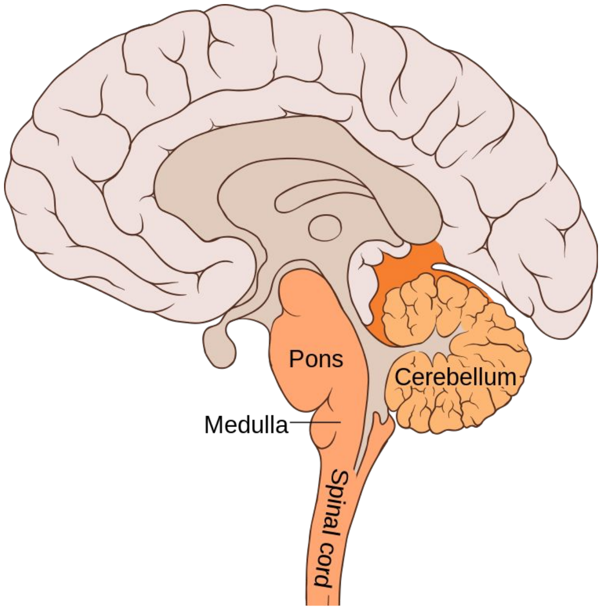
PNS Autonomic nervous system

# HindBrain

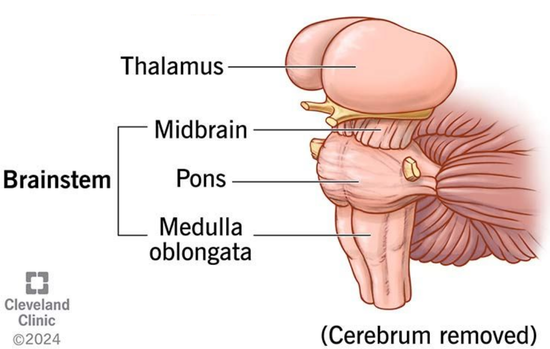
**Medulla Oblongata**
- Control primal reflexes (e.g., breathing, heart rate, coughing, vomiting)
- functions that basically keep us alive

**Pons**
- Latin for “Bridge”: relay b/w cortex & cerebellum and brain & spinal cord
- Lateral: signals to/from cerebellum
- Medial: De-/Arousal system (i.e. reticular formation for arousal, and raphe system for sleep)

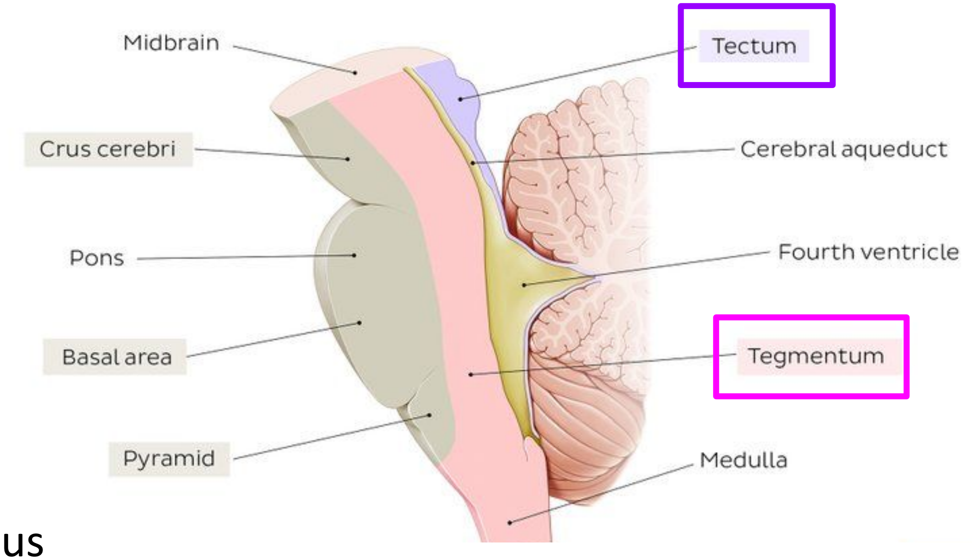
**Cerebellum**
- Store motor programs w/ real-time sensory coordination
- Critical in timing/well-coordinated actions and also important for shifting attention
- Propose motor sequences and guide movements
- NOT the brain stem




+ Hypothalamus

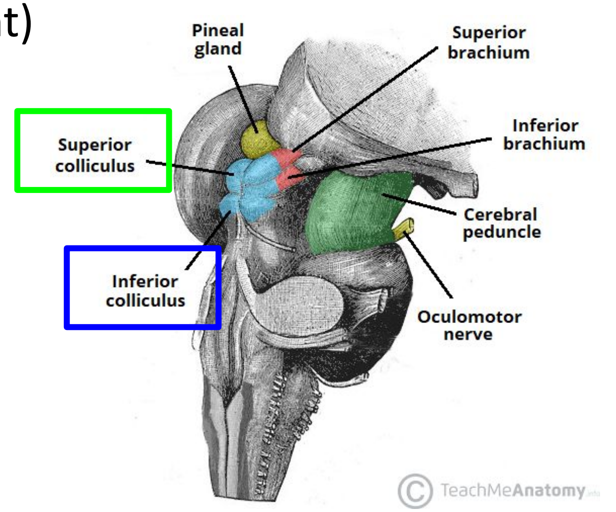


# Midbrain




**Tectum**

- Latin for "Roof"
- Part of **sensory** pathways to the brain
- Consists of superior and inferior colliculus
    1) **Superior colliculus**: visual motion (including Blindsight)
    2) **Inferior colliculus**: auditory motion



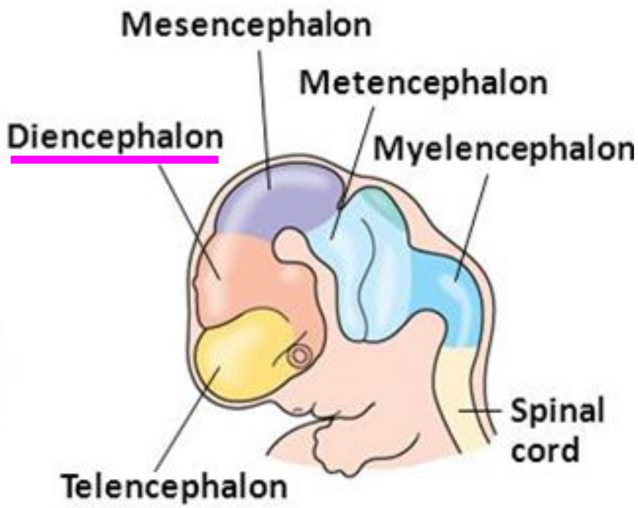

**Tegmentum**

- Latin for "Covering" or "Rug", below Tectum
- Contains major motor pathways and some cranial nerves
- Includes Red Nucleus and Substantia Nigra
- Contains cranial nerves to control eye movements

tectum to detect 'em, tegmentum for momentum

# Forebrain (Diencephalon)

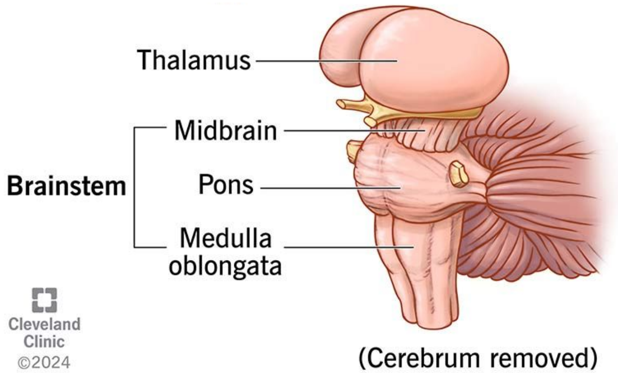
Diencephalon consists of the thalamus and hypothalamus



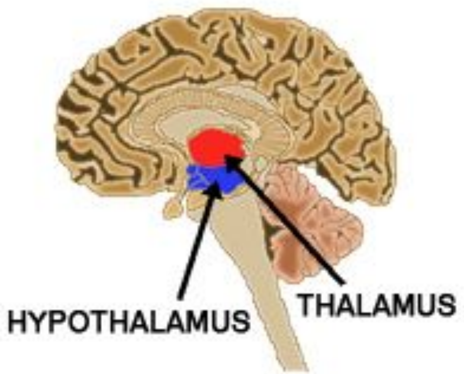

Embryo at 5 weeks

**Thalamus**
- Primary source of input to cerebral cortex
- Projects to/Receives from Sensory, Motor & Arousal sys
- Nuclei of many sensory and motor systems
- Involved in cortical arousal

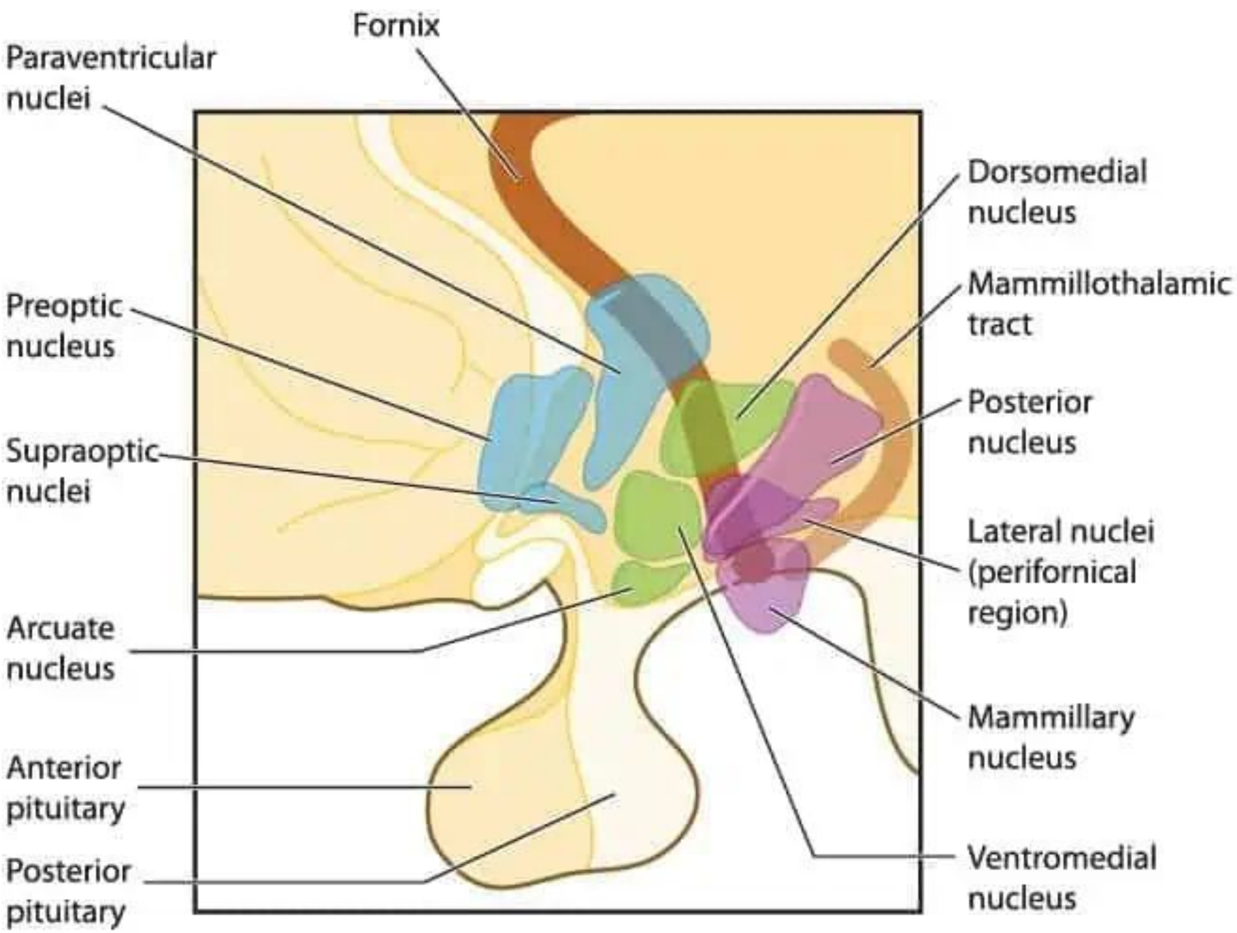
**Hypothalamus**
- Hypo = “below” ⟶ tucked in below Thalamus
- Oversees “4Fs”: Fighting, Fleeing, Feeding, F…
- Also regulates temperature and internal clock
- <u>Neuro-Endocrine</u> Sys: control brain + hormone sys by communicating to Pituitary Gland




(Cerebrum removed)



# Forebrain (Diencephalon)




nuclei of Hypothalamus

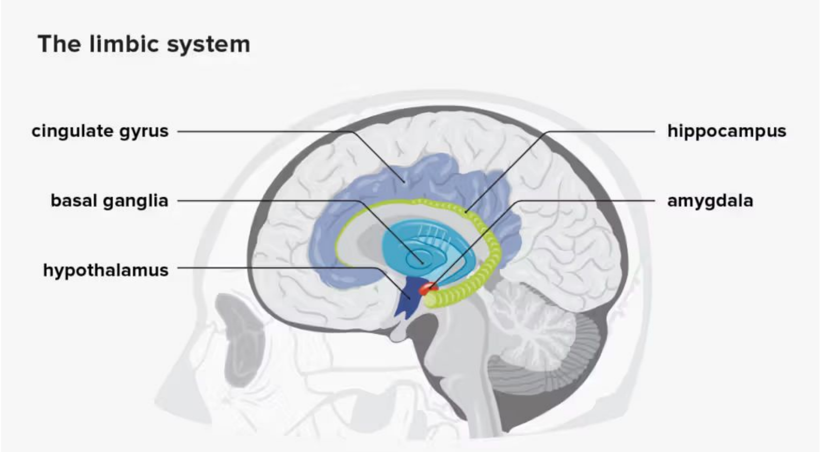
# Forebrain (Telencephalon)

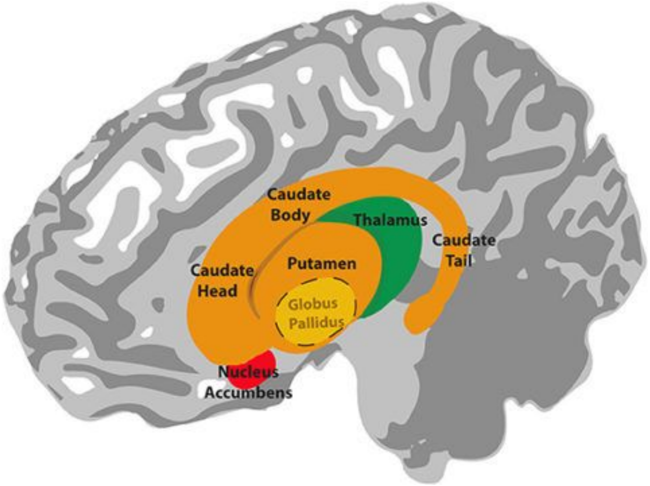
**Limbic System**

- Emotion, Motivation
- Hippocampus: formation of new memories and spatial mapping
- Amygdala: emotional expression and interpreting others’ emotions
- Cingulate Gyrus: Valence +/- Evaluator, “RE-entrant System”
- Olfactory Bulb: exchanges olfactory information with the rest of the limbic system

**Basal Ganglion**

- Includes caudate, putamen, and globus pallidus
- “Motor Area”, but diff. from cerebellum (which muscle to use): organizing activity into TASKS, especially **planned sequential behaviors** (task setting, check goals…)
- Another “RE-entrant system”: bottom-up input + hierarchical cortical analysis



# Forebrain (Telencephalon)

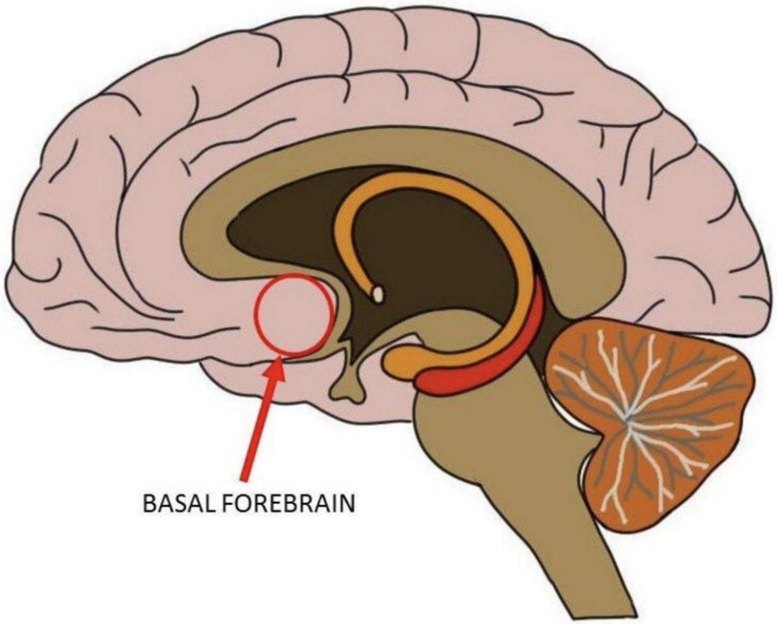
**Basal Forebrain**

- Cortical area just anterior to Hypothalamus

- Major role in attention and cortical arousal

- Main source of ACh (Acetylcholine, excitatory neural transmitter; wakes you up in the morning…) and GABA (Gamma-Aminobutyric Acid, de-arousal/inhibitory neural transmitter; shut you down, put you to sleep…)

- Receives input from Raphe/Reticular Arousal System in Brainstem



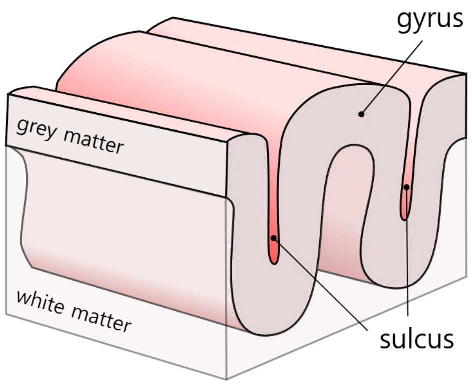
# Cerebral Cortex

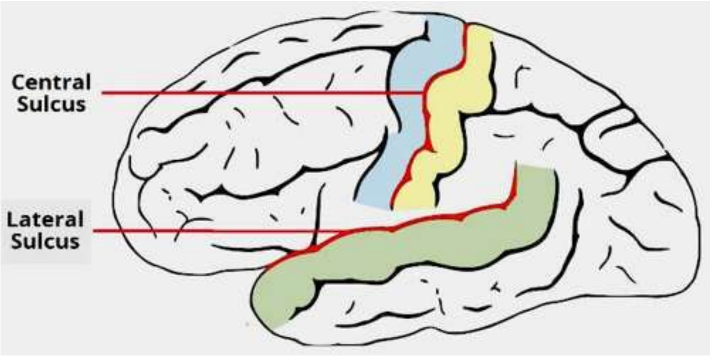
**Organized into 6 layers** (highly convoluted)

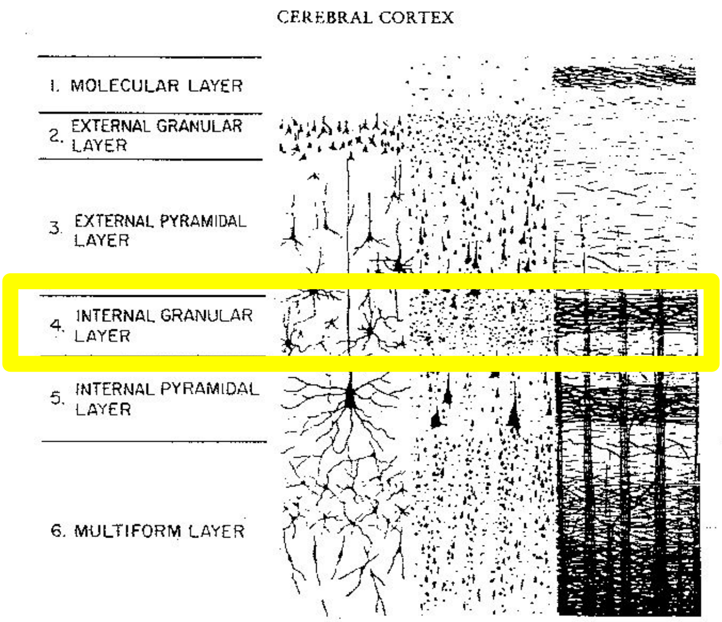
- Information projected to cortex enters at layer 4

- Bulges = gyri (sing. gyrus)
- Folds = sulci (sing. sulcus)

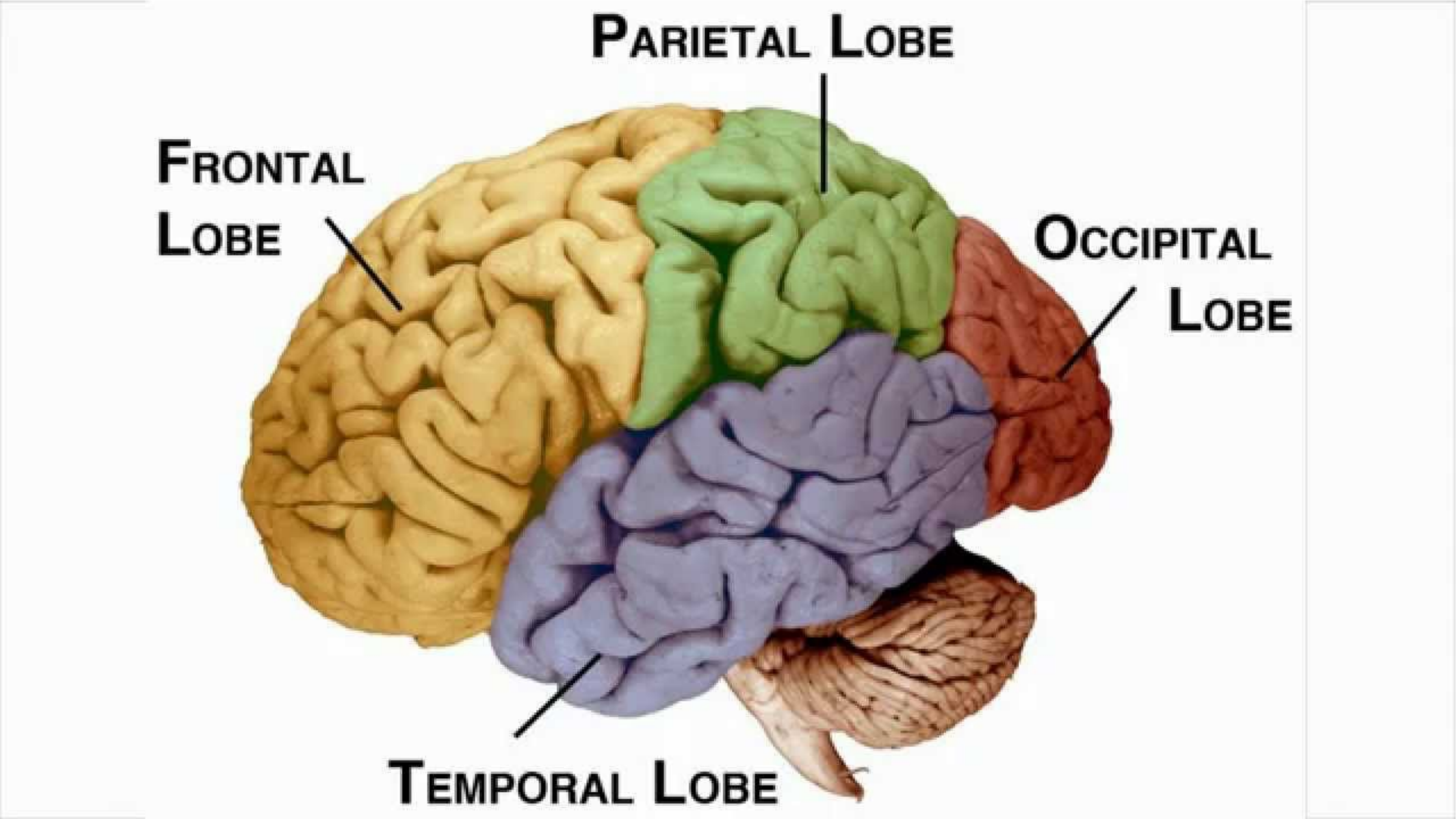
Landmarks
- Central Sulcus divides parietal from frontal lobe
- Lateral Sulcus/Fissure divides temporal from frontal lobe



# Lobes of the Brain



# Breaking down the Lobes

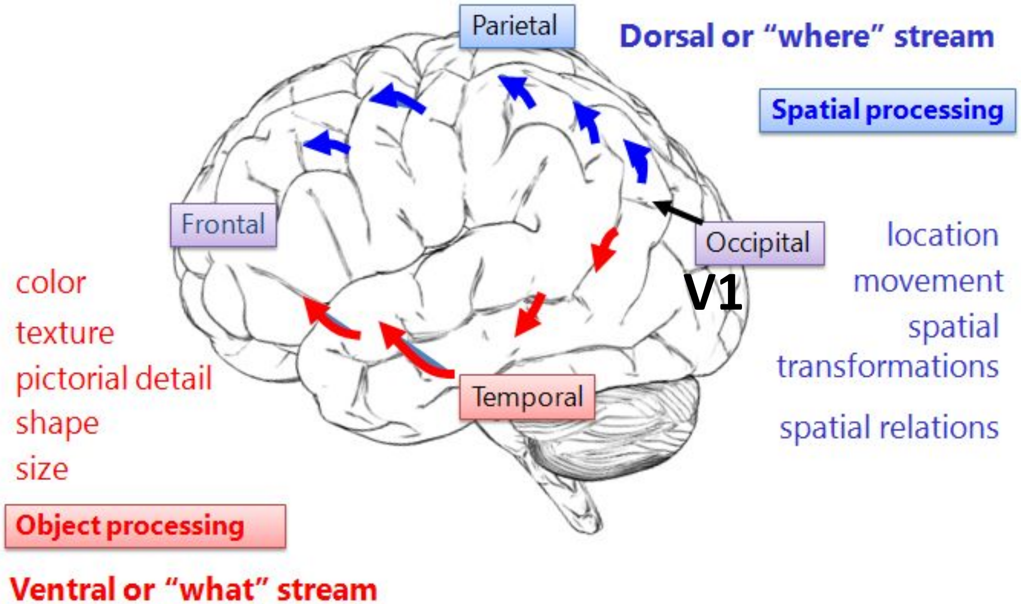
**Occipital Lobe**

- <u>Devoted</u> to Visual processing
- Contains V1 (primary visual cortex) and receives projections from thalamus
- major visual pathways: Ventral/Dorsal stream

**Temporal Lobe**

- Contains Medial Temporal (MT) - includes Direction-sensitive motion detectors, Medial Superior Temporal (MST) - includes Optic flow detectors, Ventral Visual Pathway - include Face cells,, Auditory areas - includes Wernicke's Area, and Anterior Temporal - emotional expression and interpretation (esp. right hem)

Ventral
= Who/What
: color & detail




Dorsal
Where/How
: Motion & Depth

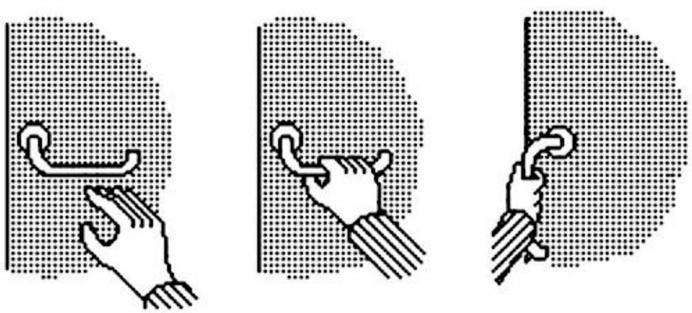
# Breaking down the Lobes

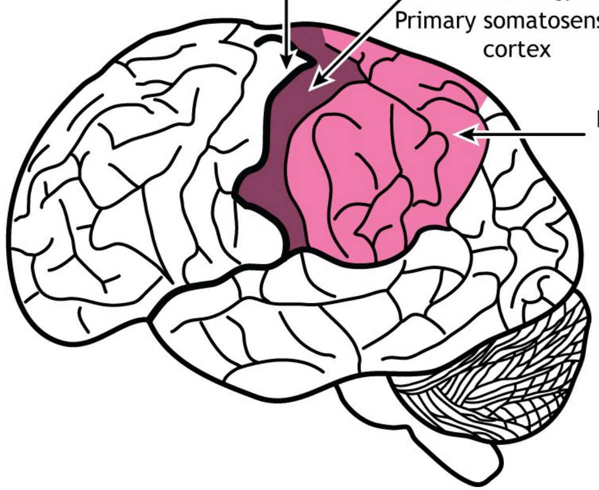
**Parietal Lobe**

- integrating visual and somatosensory info (touch, pain, temperature, proprioception such as body position and movement, etc)
- Post-central gyrus (S1): primary somatosensory cortex
- also includes higher visual areas of "where/how" pathway, and the examples are…

**Canonical cells**

- Responds to "affordances" of objects ("Can I grab this object? How?)
- Activity reverberates w/ premotor cortex, to shape how hand approaches

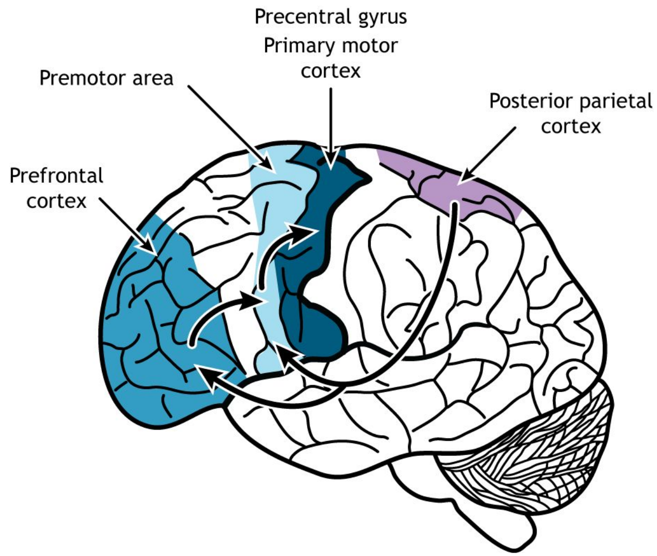


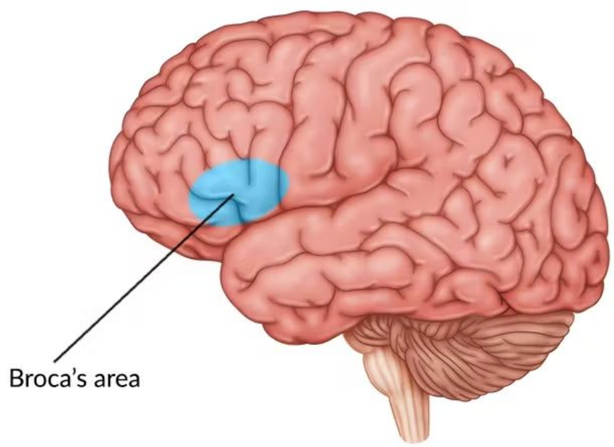


# Breaking down the Lobes

**Frontal Lobe**

- Important for motor movements, language production, and strategy
- Precentral gyrus: primary motor cortex (initiating and controlling voluntary movements, “Ok, execute!”)
- Premotor areas: anterior to motor cortex (prepare to act → planning)
    - includes mirror cells (w/ Parietal) which responds to seeing self or other perform familiar manual tasks → simulation of observed action
    - includes Broca’s Area (activate just before speech production and articulation)
- Prefrontal cortex



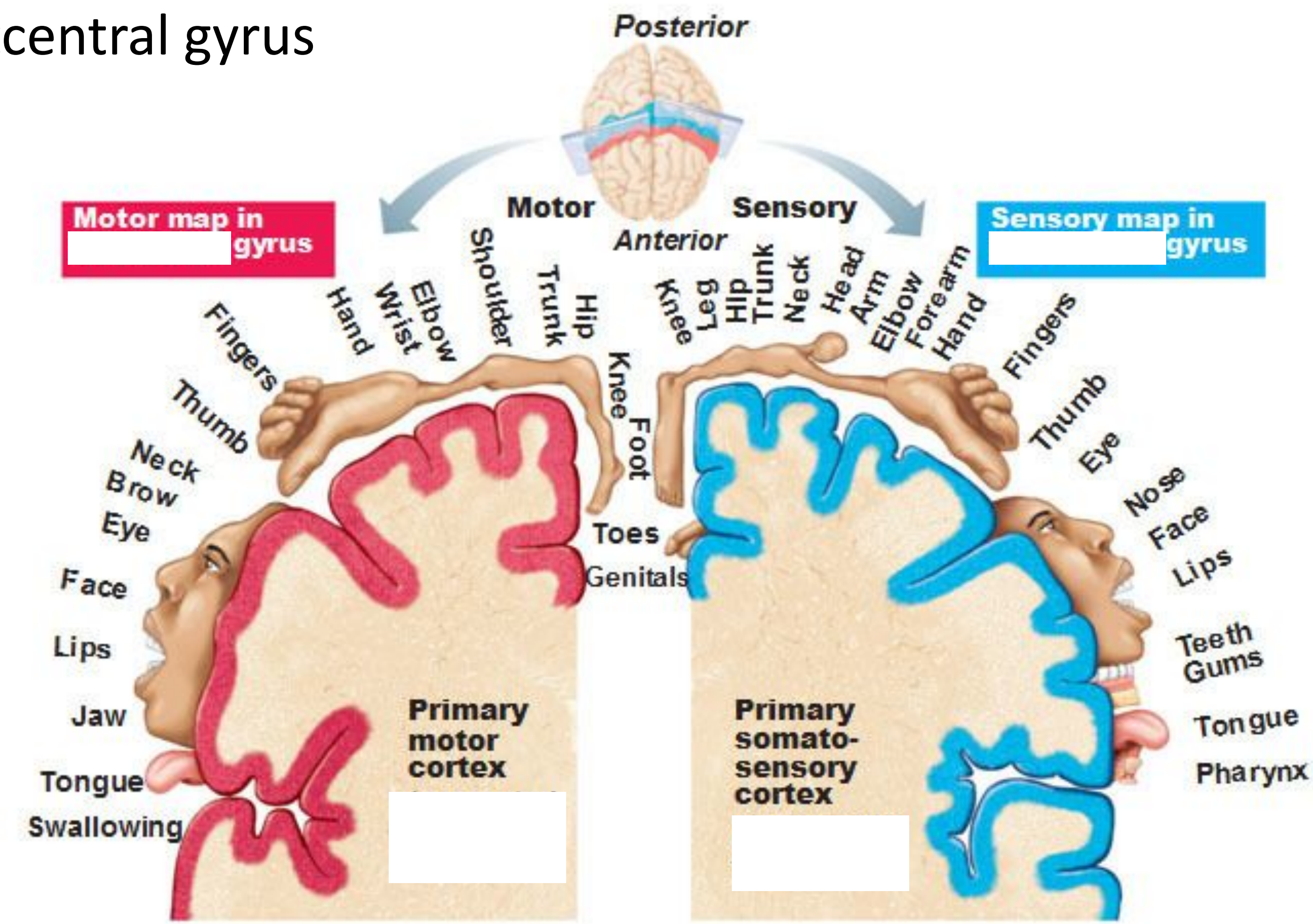



# Breaking down the Lobes

*Do you remember…?*

- precentral gyrus
- postcentral gyrus

*Note that each hemisphere receives info from the opposite side of the body*



# Breaking down the Lobes

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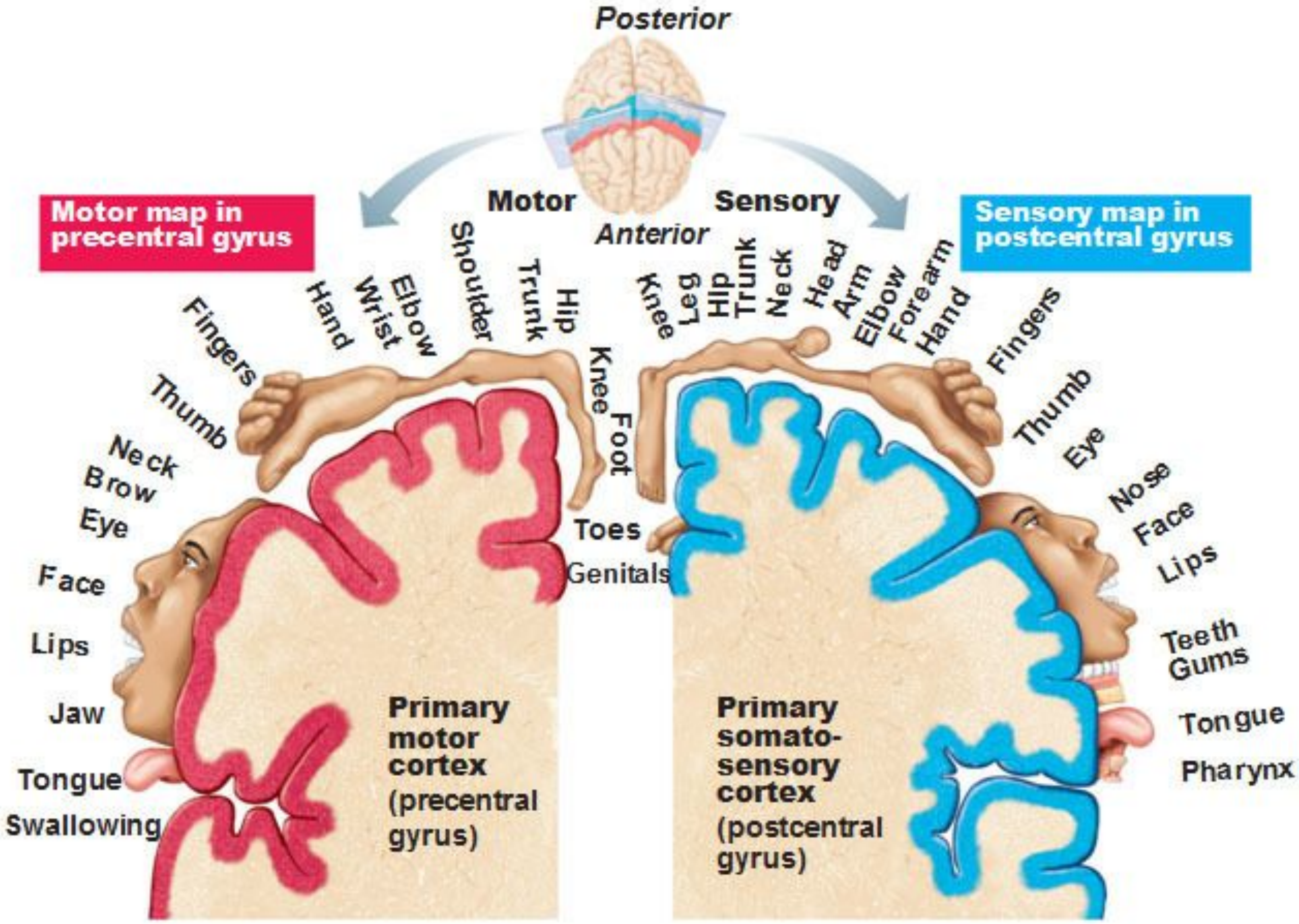
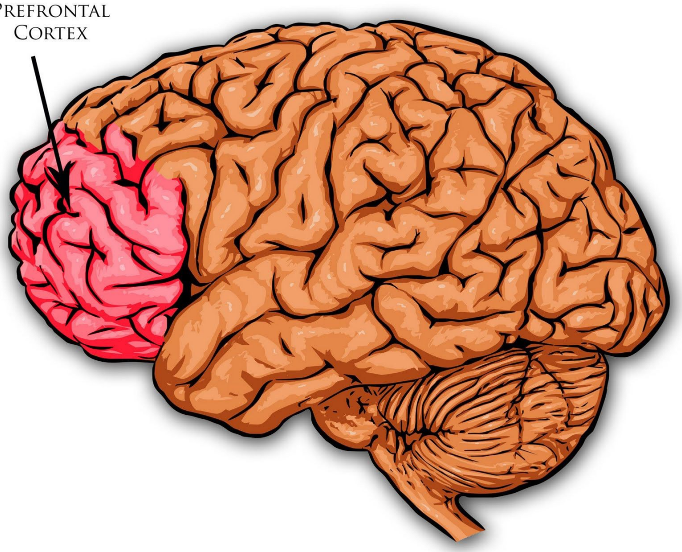
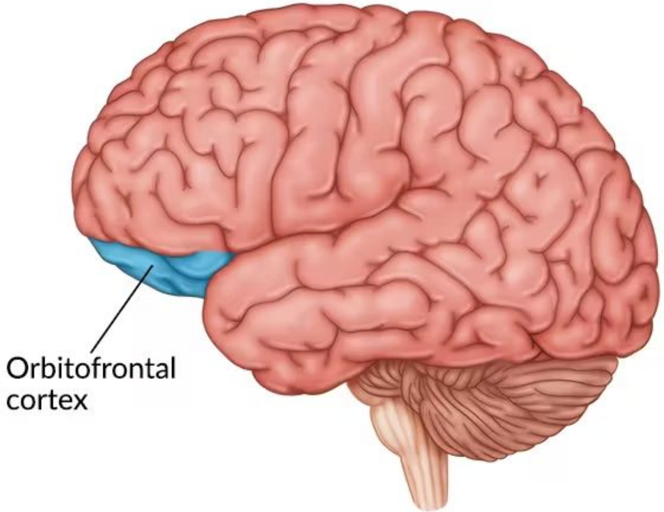


# Breaking down the Lobes

# Prefrontal Cortex

- Self-control
- Delayed gratification
- Culturally/socially appropriate behavior
- Cost/benefit analysis
- Orbital-Frontal Cortex
    - Plays an important role in evaluating behavior of self and other, social strategy

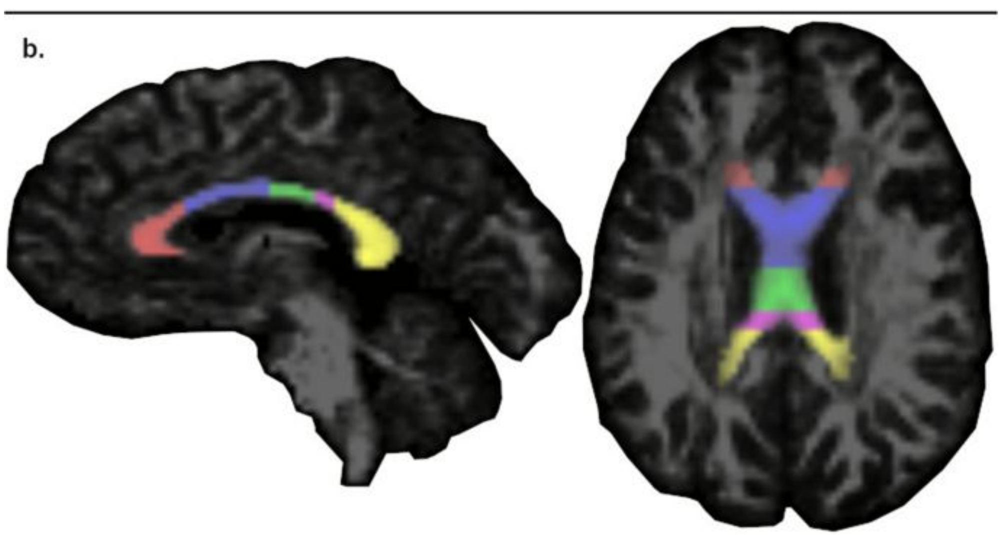


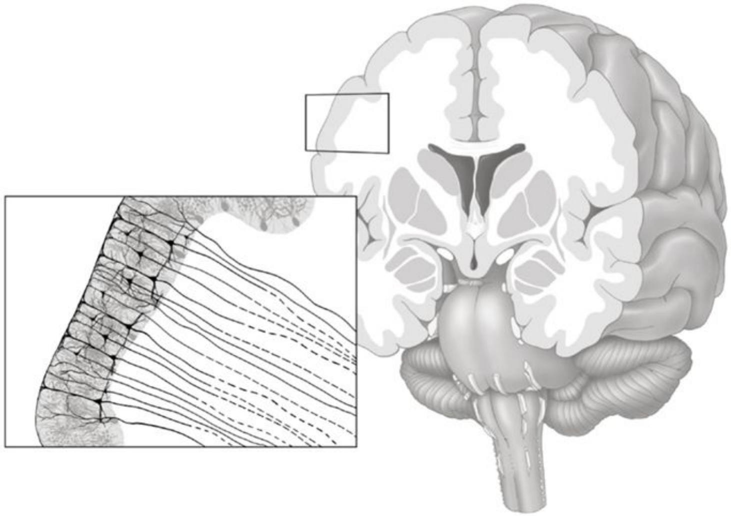



# Corpus Callosum

- Large axonal fibers connecting the two hemispheres

- Part of the “white matter” (connections b/w the little grey cells)
  - Consists of mainly myelinated axons
  - Brain = ~66% White Matter, by volume



https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2015.00035/full

# Spinal Cord

Consists of 31 segments along the spine

**Dorsal (back part) Horns**

- Afferent nerves
- Information <u>from body to brain</u>

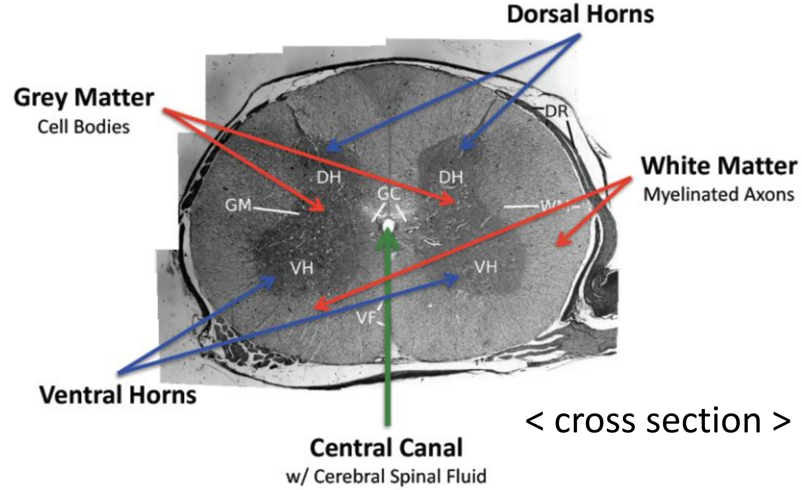
**Ventral (front part) Horns**

- Efferent nerves
- Motor information <u>to muscles and glands</u>

**Bell-Magendie Law**

- Sensory information enters (rear) dorsal horn
- motor information exits (front) ventral horn

→ *In the Door and out the Vent, that's how Spinal Cord info's sent*

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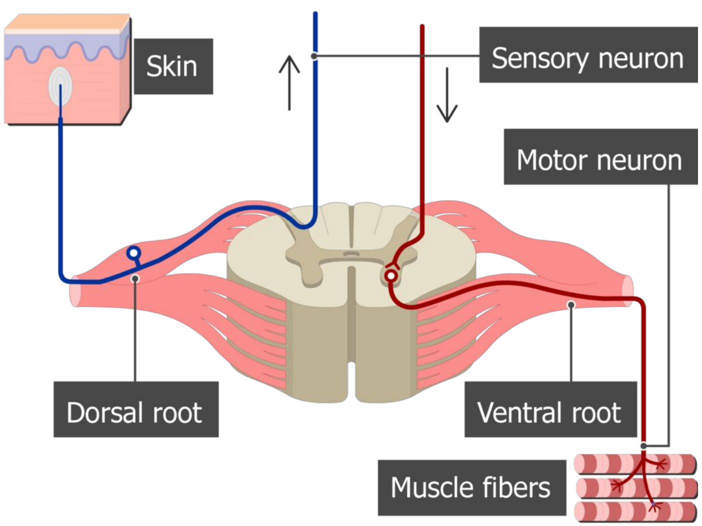


# Peripheral Nervous System

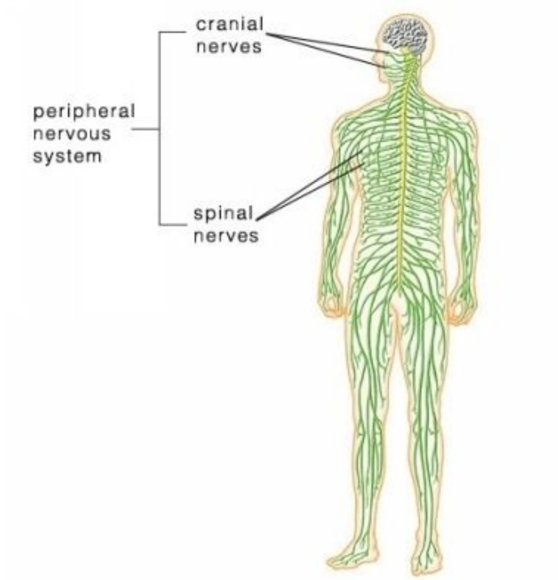
**Somatic NS**

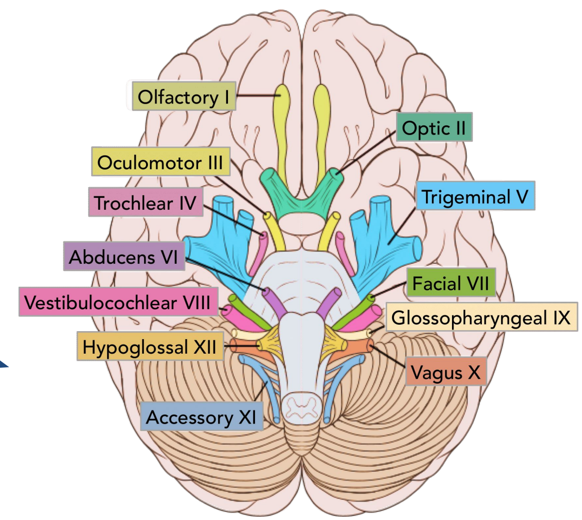
- 31 pairs of spinal nerves
    - Carry sensory input *from the* skin, muscles, and joints *to* the spinal cord.
    - Carry motor output *from* the spinal cord *to* the skeletal muscles.

  Structure of each spinal nerve pair:

    - Dorsal root (afferent): brings *sensory* signals into the spinal cord.
    - Ventral root (efferent): sends *motor* signals out to the muscles

- 12 cranial nerves: carrying sensory information from senses, receive feedback from some organs, control motor output to muscles that control eye movements, facial expressions



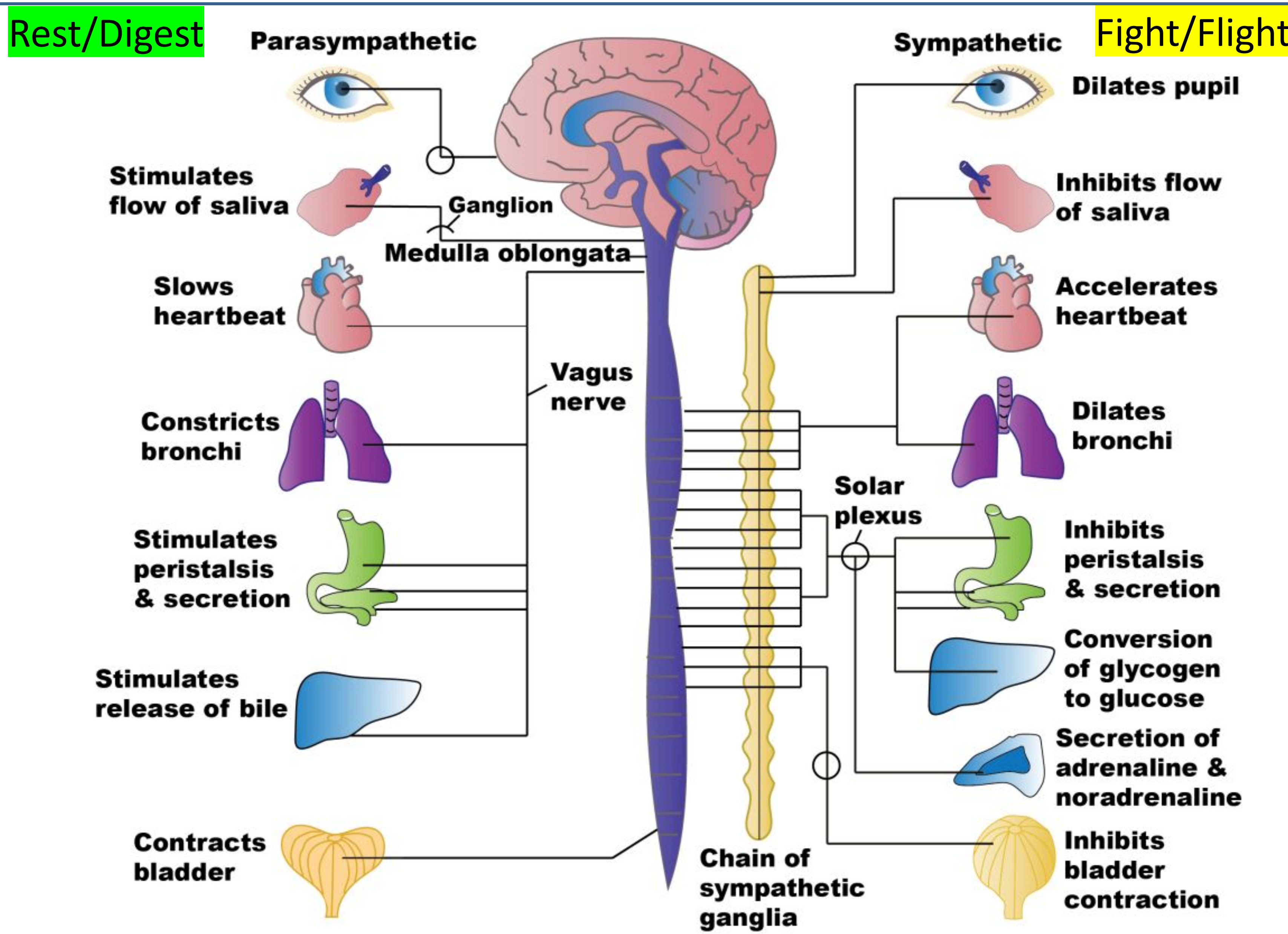




**Autonomic NS**

- Regulates internal state

1) Sympathetic "Fight or Flight"
2) Parasympathetic "Rest & Digest"

# Autonomic Nervous System



# Review Questions

*Label as either a sympathetic response (S) or a parasympathetic response (P):*

___ Pupils dilate
___ Increased heartrate
___ Saliva production
___ Bronchi constrict
___ Halt digestion
___ Facilitate sexual arousal
___ Hold bladder
___ Blood vessels constrict
___ Pilo-erection

# Review Questions

*Label as either a sympathetic response (S) or a parasympathetic response (P):*

_S_ Pupils dilate
_S_ Increased heartrate
_P_ Saliva production
_P_ Bronchi constrict
_S_ Halt digestion
_P_ Facilitate sexual arousal
_S_ Hold bladder
_S_ Blood vessels constrict
_S_ Pilo-erection

# See you next week!